

INTRODUCTION TO ECONOMICS

MGT404

Elasticity

Yale

Recap and Today's Plan

Last Lecture

- Market Equilibrium
- Consumer surplus and producer surplus
- When are shifts in demand or supply translated into prices versus quantities?

This Lecture

- Elasticity of Demand
- Cross-Price Elasticity
- Income Elasticity

Today's Plan

Equilibrium and elasticity:

1. Own Price Elasticity

- Own price elasticity: always negative for downward sloping demand.

2. Cross-Price Elasticity

- Complements and Substitutes.

3. Income Elasticity

- Normal goods vs inferior goods.

Motivation: Demand Elasticity

Situation in 2008

- Jan 2, 2008: Crude oil hits (then!) record \$100 per barrel; unleaded fuel \$3.05/gal (average gas price winter 2007, \$2.11)
- Apr 21, 2008: Gas hits \$3.50/gal
- May 15, 2008: Gas nears \$4/gal
- May 21, 2008: Crude hits \$130/barrel
- Jul 7, 2008: Crude peaks at \$147/barrel, while gas average is \$4.11/gal

Huge swings in gas prices:

- What was the effect on quantity in the primary market?
- What was the effect on quantity in incidental markets?

It would be great to be able to compare these effects against one another.

Demand Elasticity

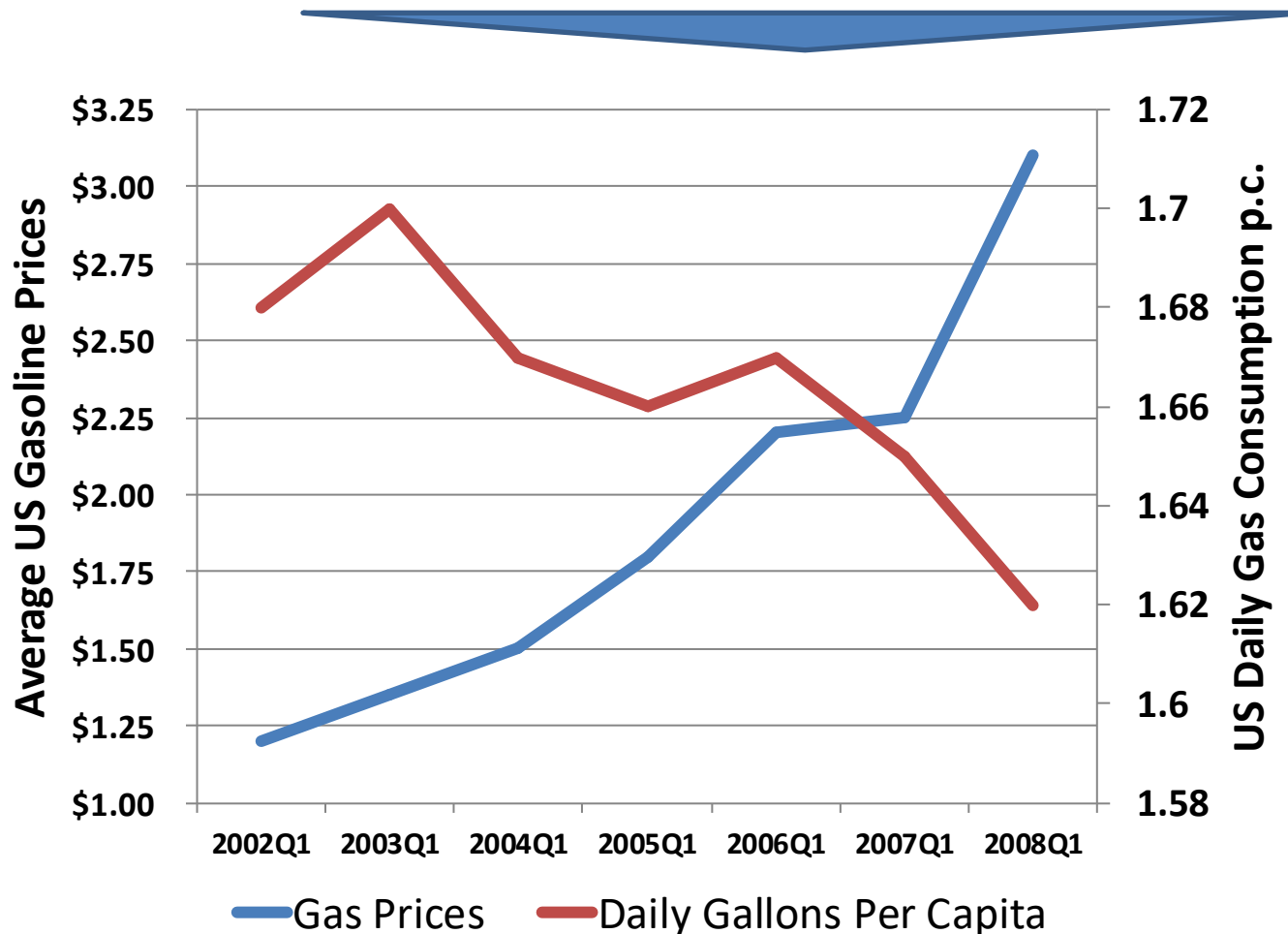
Setting

- Let's consider how the demand for gasoline and the demand for gas guzzling cars responds to changes in the price of gasoline



What do gas prices look like: 2002 to 2008

Movement in gasoline prices is due primarily to supply shifts (e.g. major hurricanes)¹



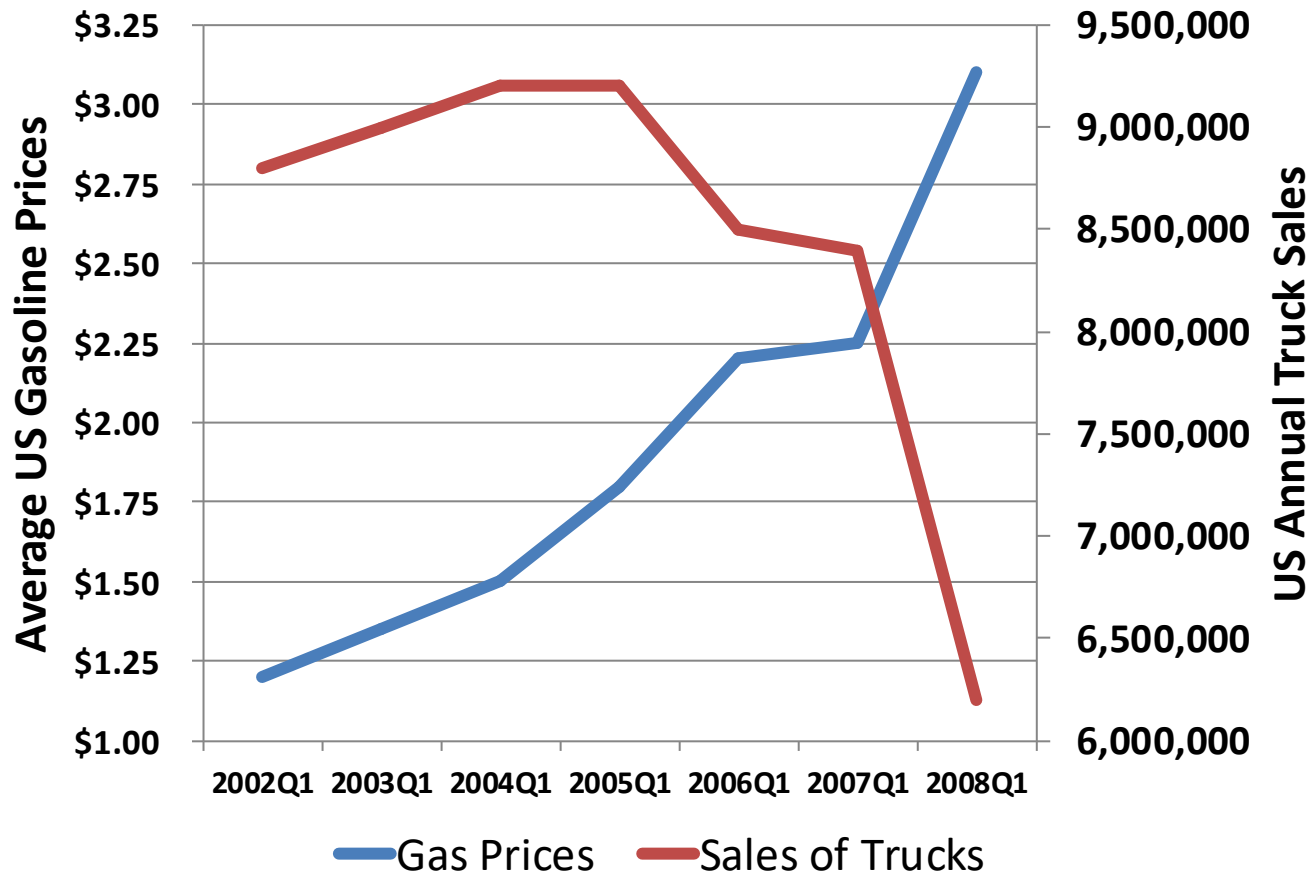
How would you describe the demand response?

- Consumption change of $1.62 - 1.68 = -0.06$ gallons p.c./day
- Price change of $\$3.10 - \$1.20 = \$1.90$

So, the rate of change is around $-0.06 / 1.90 = -0.032$ in units of "gallon per capita per day / dollar"

What do truck sales look like: 2002 to 2008

Trucks (including SUVs) consume more gas per mile driven than other vehicles



How would you describe the demand response?

- Truck sales change by 6.2M-8.8M=-2.6M
- Price change of \$3.10-\$1.20=\$1.90

So, the rate of change is around $-2.6/1.90 = -1.37$ in units of "million trucks / dollar"

Problem: Comparisons

Effect on Gasoline Sales

- -0.032 in units of “gallon per capita per day / dollar”

Effect on Truck Sales

- -1.37 in units of “million trucks per year / dollar”



Which one is bigger? Which one matters more?

Ideally, we would have a way of comparing that didn't depend on units...

Elasticity accomplishes this by using percentage changes:

$$\frac{\text{Percent change in } Q}{\text{Percent change in } P}$$

Elasticity as a General Concept

How responsive is...

Crime, to minimum sentences?

On time class arrival, to class start time?

Number of Job applications, to starting salary?



Elasticity is a concept for measuring how responsive a variable is to a change in another.

What is the *percent change* in one variable for a *one percent change* in another?

Elasticity

We want to compare the responses in gas consumption and truck sales from a change in the price of gas:

Effect on Gasoline Sales

- -0.032 in units of “gallon per capita per day / dollar”

Effect on Truck Sales

- -1.37 in units of “million trucks per year / dollar”

Percentage Effect on Gasoline Sales

- Percentage change in gas price: $\frac{\Delta P}{P}$
 $= \frac{\$3.10 - \$1.20}{\$1.20} = 158\%$
- Pct change in gas consumption:
 $\frac{\Delta Q_{gas}}{Q_{gas}} = \frac{1.62 \text{ gal} - 1.68 \text{ gal}}{1.68 \text{ gal}} = -4\%.$

Percentage Effect on Truck Sales

- Percentage change in gas price: $\frac{\Delta P}{P}$
 $= \frac{\$3.10 - \$1.20}{\$1.20} = 158\%$
- Pct change in truck consumption:
 $\frac{6.2\text{M trucks} - 8.8\text{M trucks}}{8.8\text{M trucks}} = -30\%.$

Elasticity of gasoline w.r.t. gasoline price:

$$\frac{-4\%}{158\%} = -\mathbf{0.025}.$$

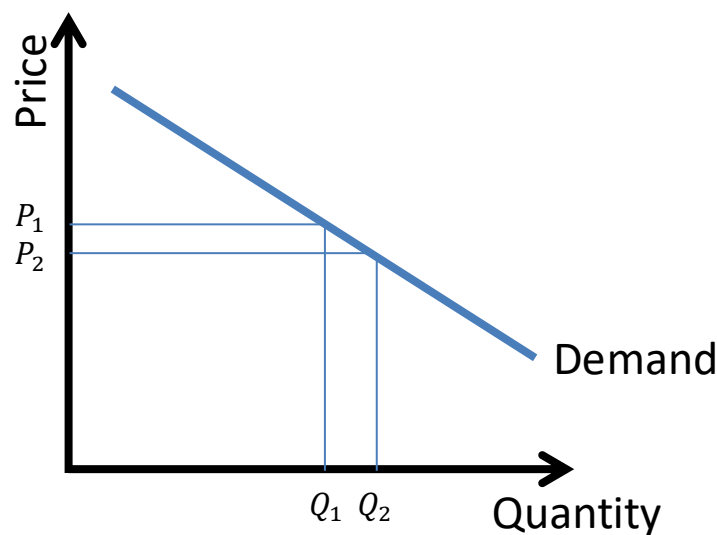
Elasticity of trucks w.r.t. gasoline price:

$$\frac{-30\%}{158\%} = -\mathbf{0.190}.$$

Own-Price Elasticity of Demand

Definition: Own-Price Elasticity of Demand

Elasticity is the **percentage change in quantity for each percentage change in price** for a small movement around a point on the curve.



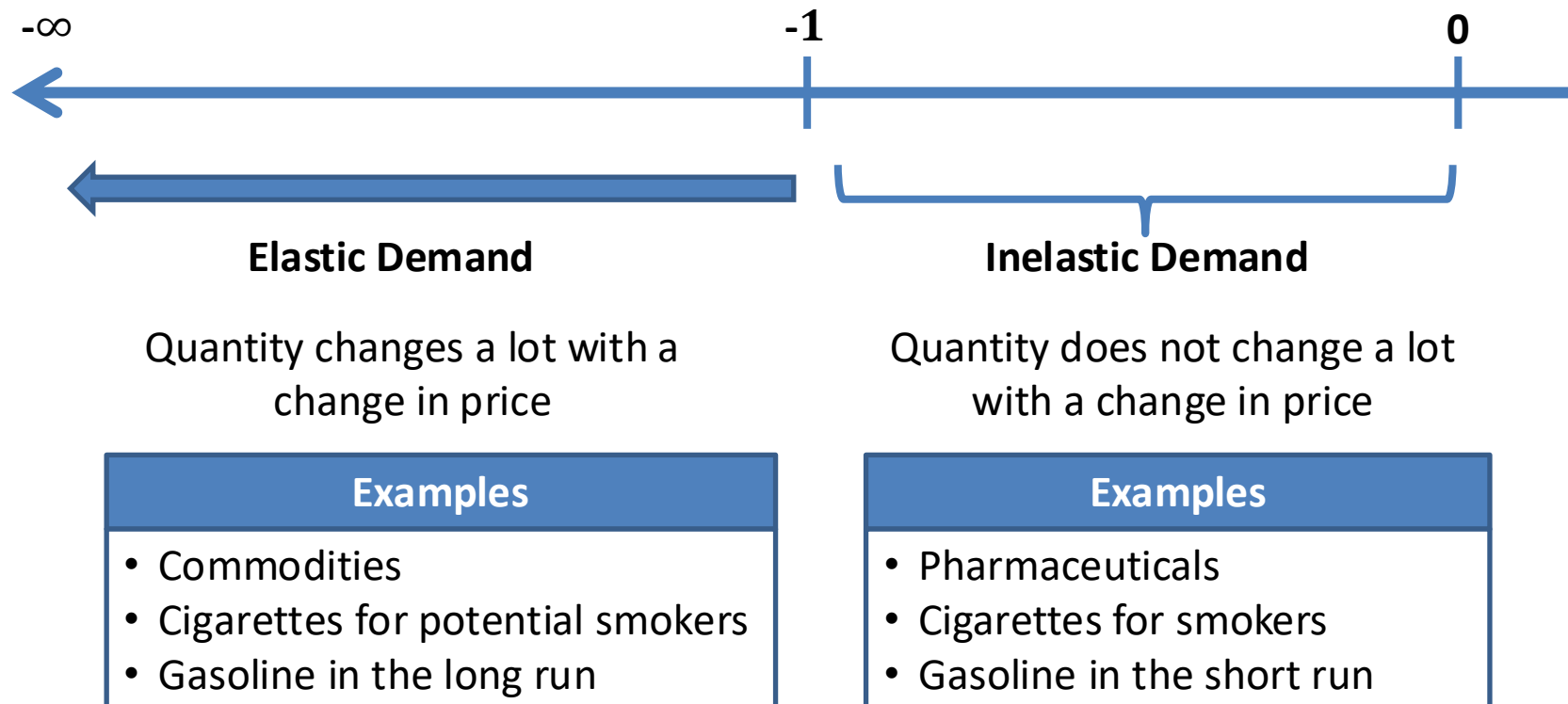
$$\text{Percentage change in quantity} = \frac{Q_2 - Q_1}{Q_1} = \frac{\Delta Q}{Q_1}$$

$$\text{Percentage change in price} = \frac{P_2 - P_1}{P_1} = \frac{\Delta P}{P_1}$$

$$\text{Elasticity} = \frac{\frac{\Delta Q}{Q_1}}{\frac{\Delta P}{P_1}} = \frac{\Delta Q}{\Delta P} \cdot \frac{P_1}{Q_1}$$

$$\text{Own-Price Elasticity of Demand: } \varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q} \approx \frac{\frac{\Delta Q}{Q_1}}{\frac{\Delta P}{P_1}} = \frac{\Delta Q}{\Delta P} \cdot \frac{P_1}{Q_1}$$

Ranges of Own-Price Elasticity



What impacts elasticity in general?

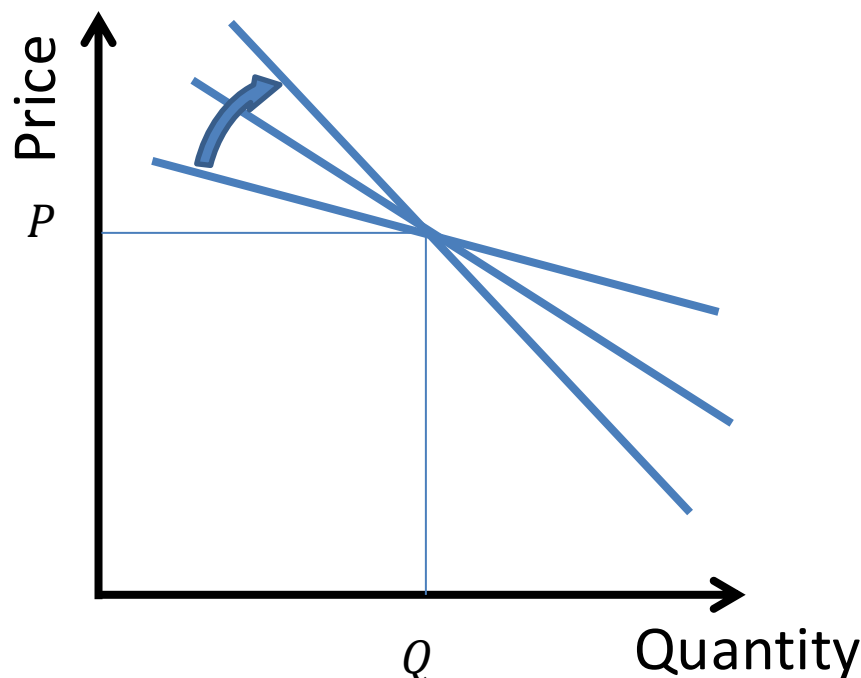
- Time: **demand tends to be more inelastic in the short run, in long run you substitute**
- Specificity: demand tends to be more elastic for more specific categories
- Market Definition: your product demand can be more or less elastic depending on how the market is defined.

Demand Elasticity Varies with Slope

What happens to Elasticity as the slope of a demand curve changes?

Rotate a demand curve around a point:

$$\text{Recall: } \varepsilon = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$



We're holding $\frac{P}{Q}$ fixed, but $\frac{\Delta Q}{\Delta P}$ gets smaller as the line becomes steeper.

- We think of flatter demand curves as being “more elastic” (think: quantity changes a lot compared to price)
- We think of steeper demand curves as being “more inelastic” (think: big changes in quantity produce small changes in price)
- Elasticity also changes *along* a demand curve.

Example: Computing Demand Elasticity

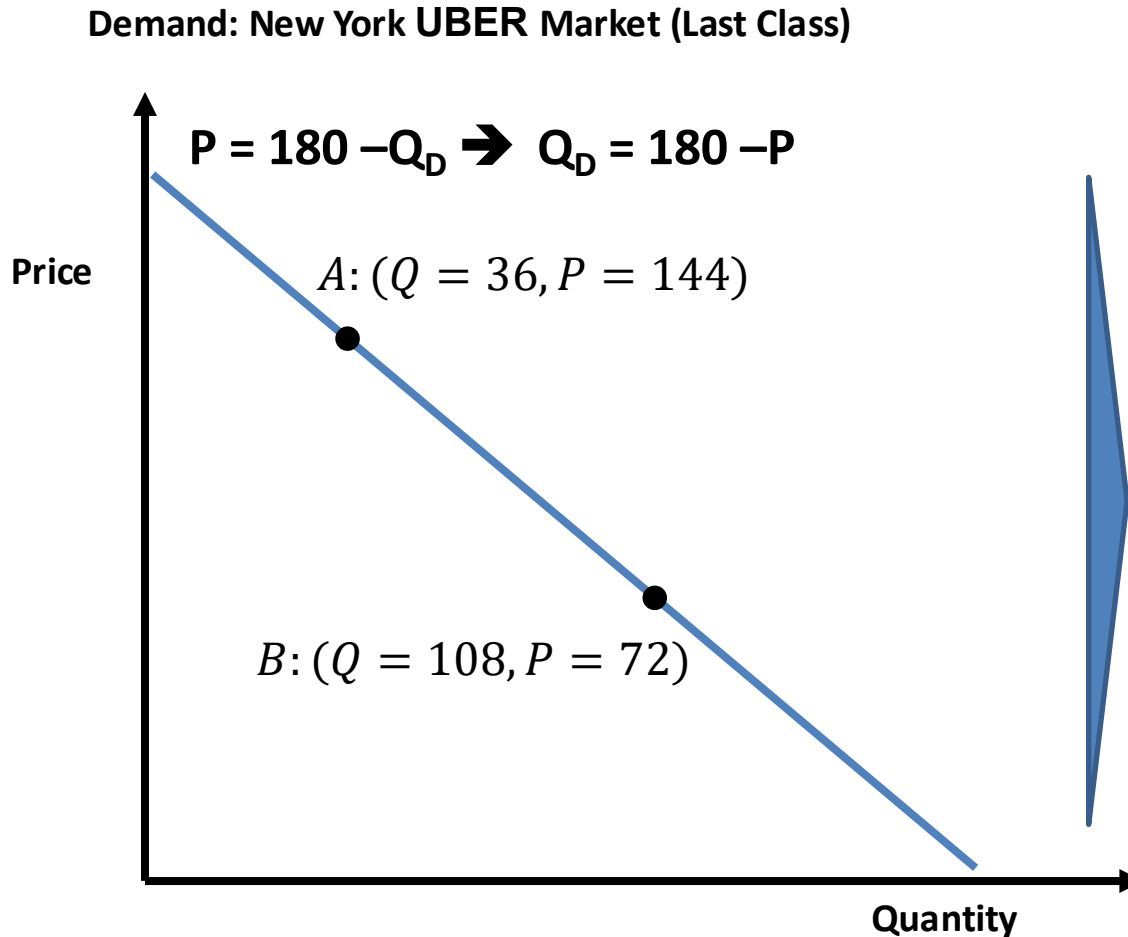
$$\text{Recall: } \varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q}$$

Elasticity at A is computed as:

$$\begin{aligned}\varepsilon &= \frac{dQ}{dP} \cdot \frac{P}{Q} \\ &= (-1) \cdot \frac{144}{36} \\ &= -4\end{aligned}$$

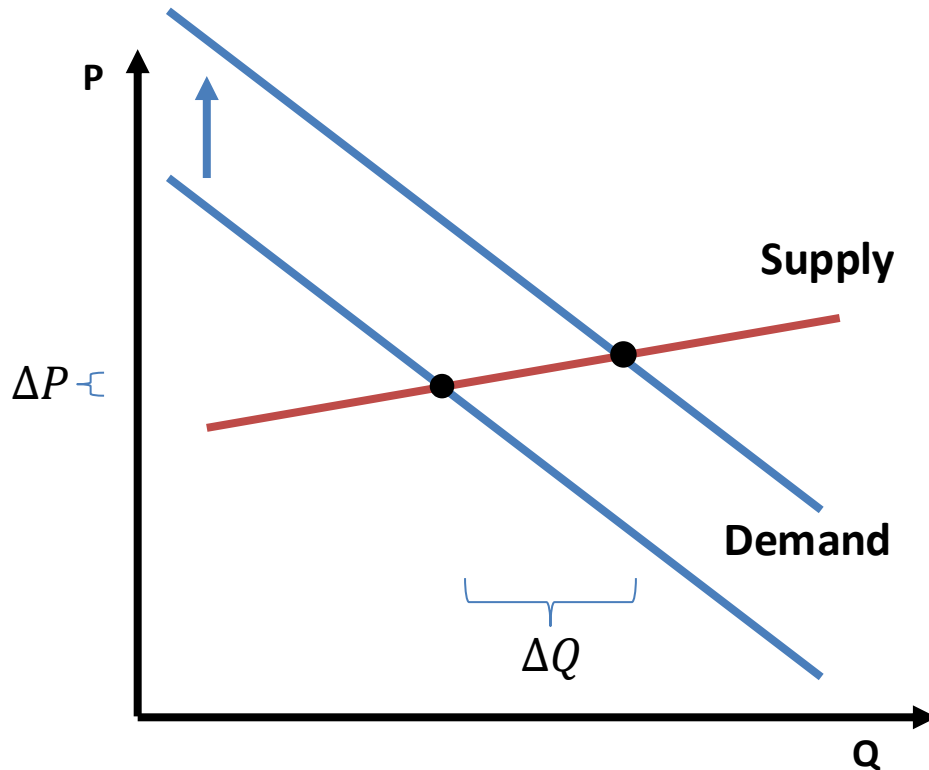
Elasticity at B is computed as:

$$\begin{aligned}\varepsilon &= \frac{dQ}{dP} \cdot \frac{P}{Q} \\ &= (-1) \cdot \frac{72}{108} \\ &= -0.67\end{aligned}$$



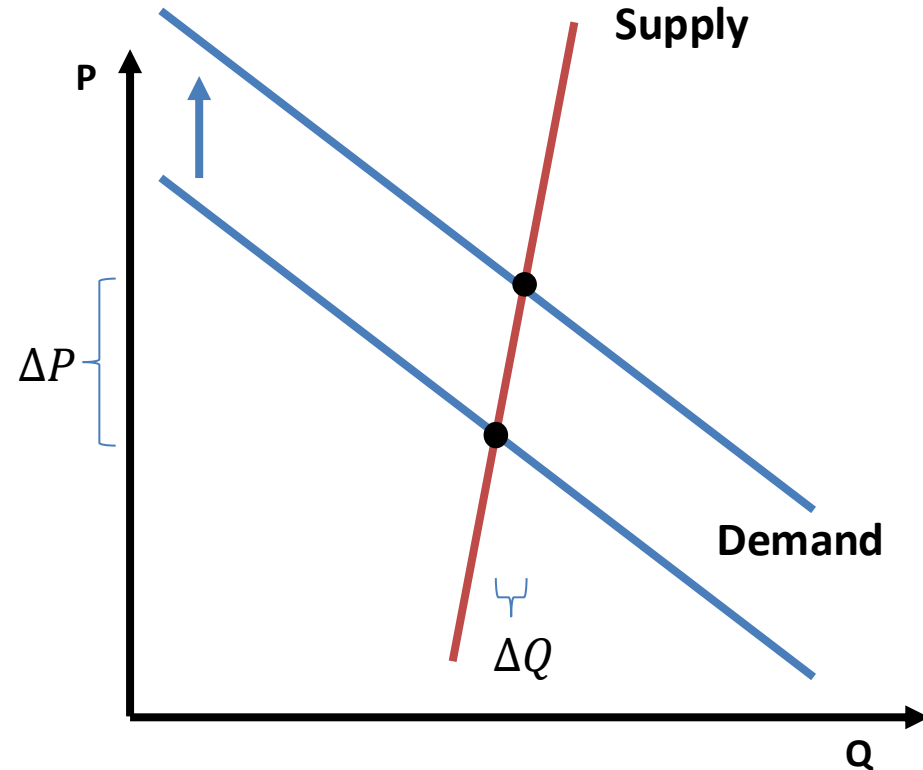
Elasticity and Shifts in Demand/Supply

Consider two cases where demand “shifts out”:



Case 1: Relatively elastic supply:

- Larger effect on quantity.
- Smaller effect on price.



Case 2: Relatively inelastic supply:

- Larger effect on price.
- Smaller effect on quantity.

Today's Plan

Equilibrium and elasticity:

1. Own Price Elasticity

- Own price elasticity: always negative for downward sloping demand.

2. Cross-Price Elasticity

- Complements and Substitutes.

3. Income Elasticity

- Normal goods vs inferior goods.

Cross Price Elasticity

Recall our gasoline and truck example from earlier:

Elasticity of gasoline w.r.t. gasoline price

- $-4\% / 158\% = -0.025$

This elasticity compares the demand for gasoline to its own price.

This is the own-price elasticity for gasoline.

Elasticity of trucks w.r.t. gasoline price

- $-30\% / 158\% = -0.190$

This elasticity compares the demand for trucks to the gas price.

This is the cross-price elasticity for trucks and gasoline.

Definition: Cross-Price Elasticity of Demand

Cross-Price Elasticity is the **percentage change in quantity for a percentage change in the price of a different product**

$$\varepsilon_{CP} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P_{other}}{P_{other}}} = \frac{\Delta Q}{\Delta P_{other}} \cdot \frac{P_{other}}{Q}$$

Cross Price Elasticity: Examples

Own-Price Elasticity



Always negative (law of demand).

Cross-Price Elasticity



Can be positive or negative.

Cross-Price Examples:

Elasticity of demand for...	... w.r.t price of...	 Positive or negative?
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- Coffee
- iPhone
- Margarine

- Tea
- Android
- Butter

....

- Pencils
- Wine
- Video Games

- Erasers
- Cheese
- Video game consoles

- Positive
- Positive
- Positive

- Negative
- Negative
- Negative

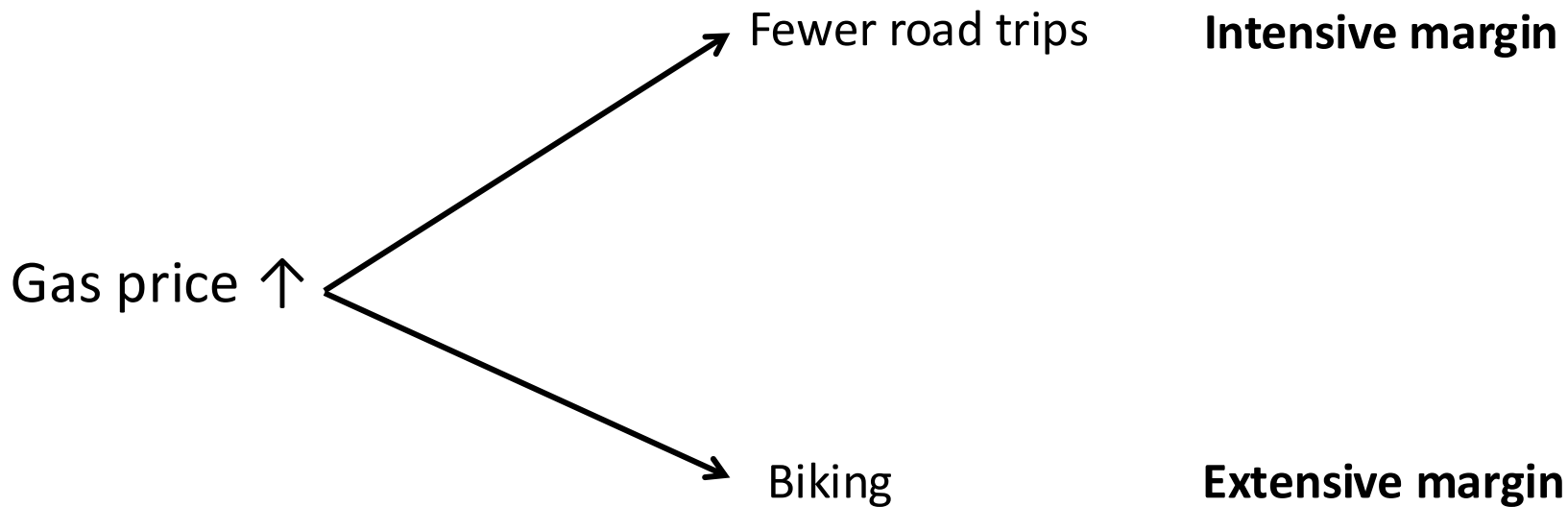


Substitute goods have **positive** cross-price elasticities.



Complementary goods have **negative** cross-price elasticities.

Where do consumers go after the price goes up?



What are other examples?

How about voting?

The Median Voter Theorem

- **Intellectual history:** Hotelling (1929) studies a model of competition between two firms offering horizontally differentiated products.
- Two ice cream cars on a beach, where do they stand?
 - **Answer:** In the middle, since people always substitutes away from the one that's too far to the one that's closer.
- Downs (1957): Same goes for voting!
- **Caveat: Voters substitute both on the extensive and the intensive margin!**

The Median Voter Theorem

Whenever two office-seeking candidates compete for a (one-dimensional) political spectrum, they always stand at the position of the median voter.



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Income Effect

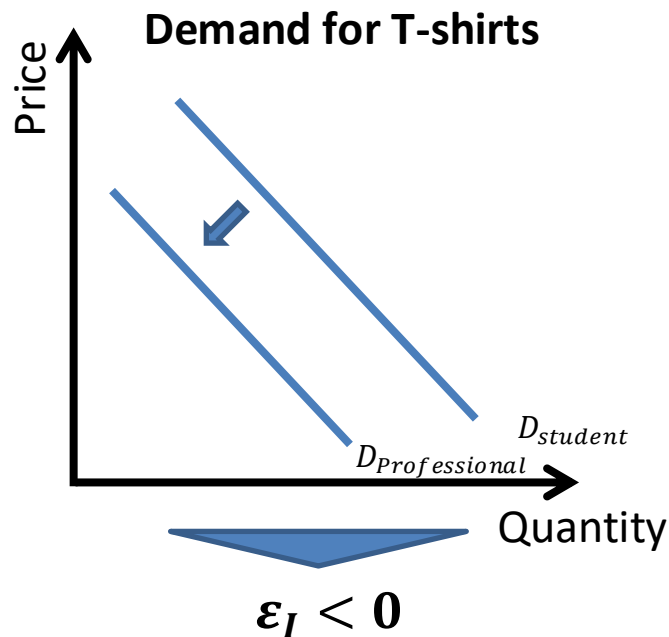
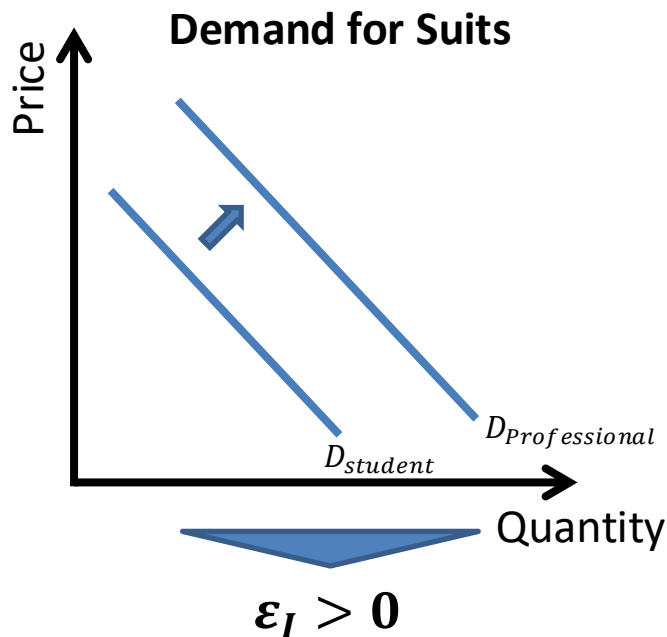
Suppose prices for a good remain the same, but people have more money to spend:

Definition: Income Elasticity of Demand

Income Elasticity of Demand is the **percent change in quantity demanded for a percent change in income**

$$\varepsilon_I = \frac{dQ}{dI} \cdot \frac{I}{Q}$$

Will income elasticity be always positive or negative?



Ranges for Income Elasticity

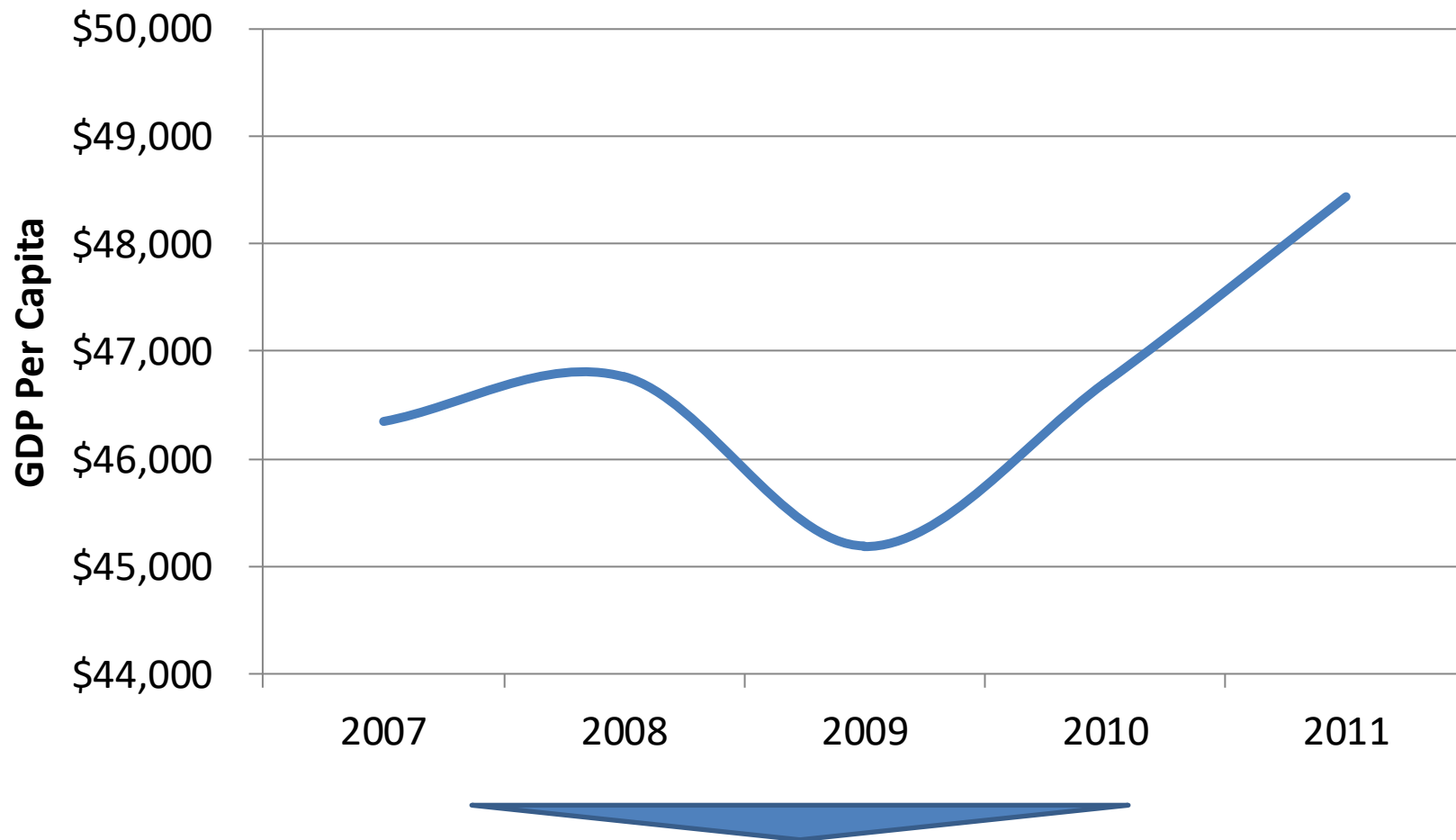
Examples of Income Elasticity:

Income Elasticity of demand for...	Positive or negative?	
• Public transit • Frozen dinners • Ramen noodles	• $\varepsilon_I < 0$ • $\varepsilon_I < 0$ • $\varepsilon_I < 0$	 Inferior goods have negative income elasticities.
....		
• Cell phone • Television • Cars	• $\varepsilon_I > 0$ • $\varepsilon_I > 0$ • $\varepsilon_I > 0$	
....		
• Holidays • Jewelry	• $\varepsilon_I > 1$ • $\varepsilon_I > 1$	 Luxury/Superior goods have income elasticities greater than 1.

We can use changes in incomes during the financial crisis to determine what types of goods certain products are.

Income Trends around the Financial Crisis

United States GDP per Capita, 2007-2011



We should expect consumption of inferior goods to have grown in 2009, and consumption of normal/superior goods to have fallen.

Income Elasticity

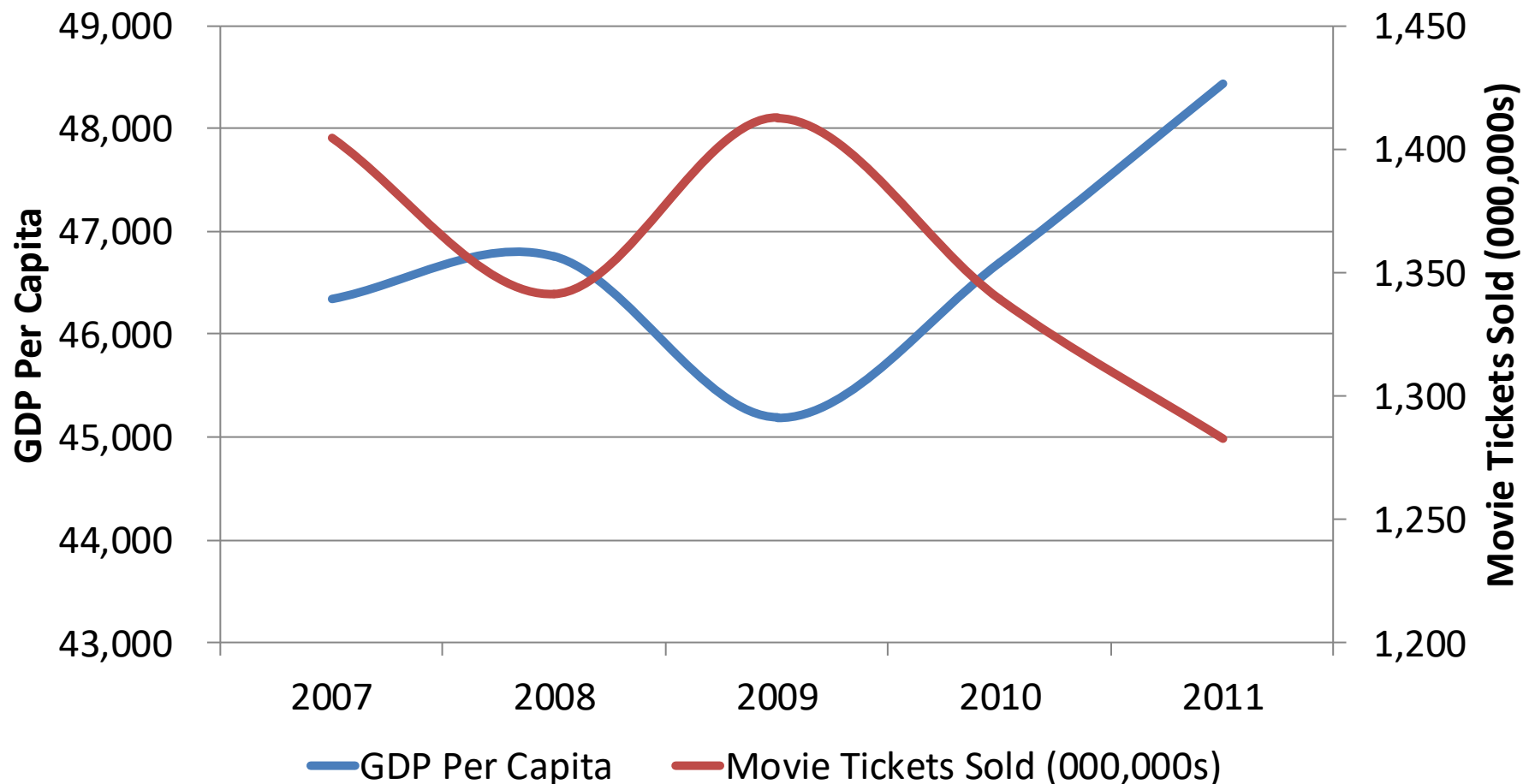
- What do we think the income elasticity of movies will be?



Income Elasticity

- What is the income elasticity of movie tickets?

The income elasticity is negative (movies are inferior)



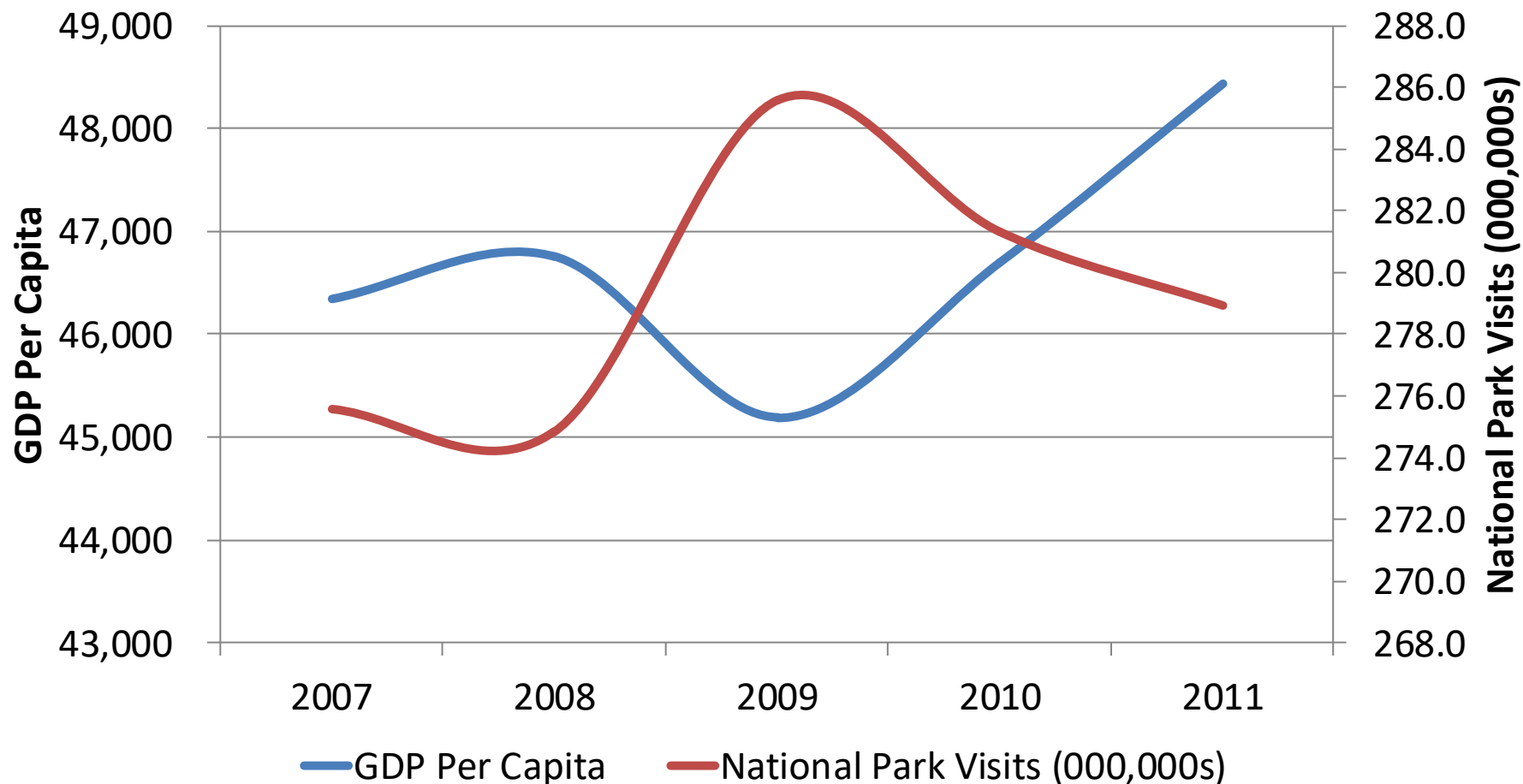
Income Elasticity

- How about visits to the National Parks?



Income Elasticity

- What is the income elasticity of national parks visits?
 - Also negative (national park visits are inferior)



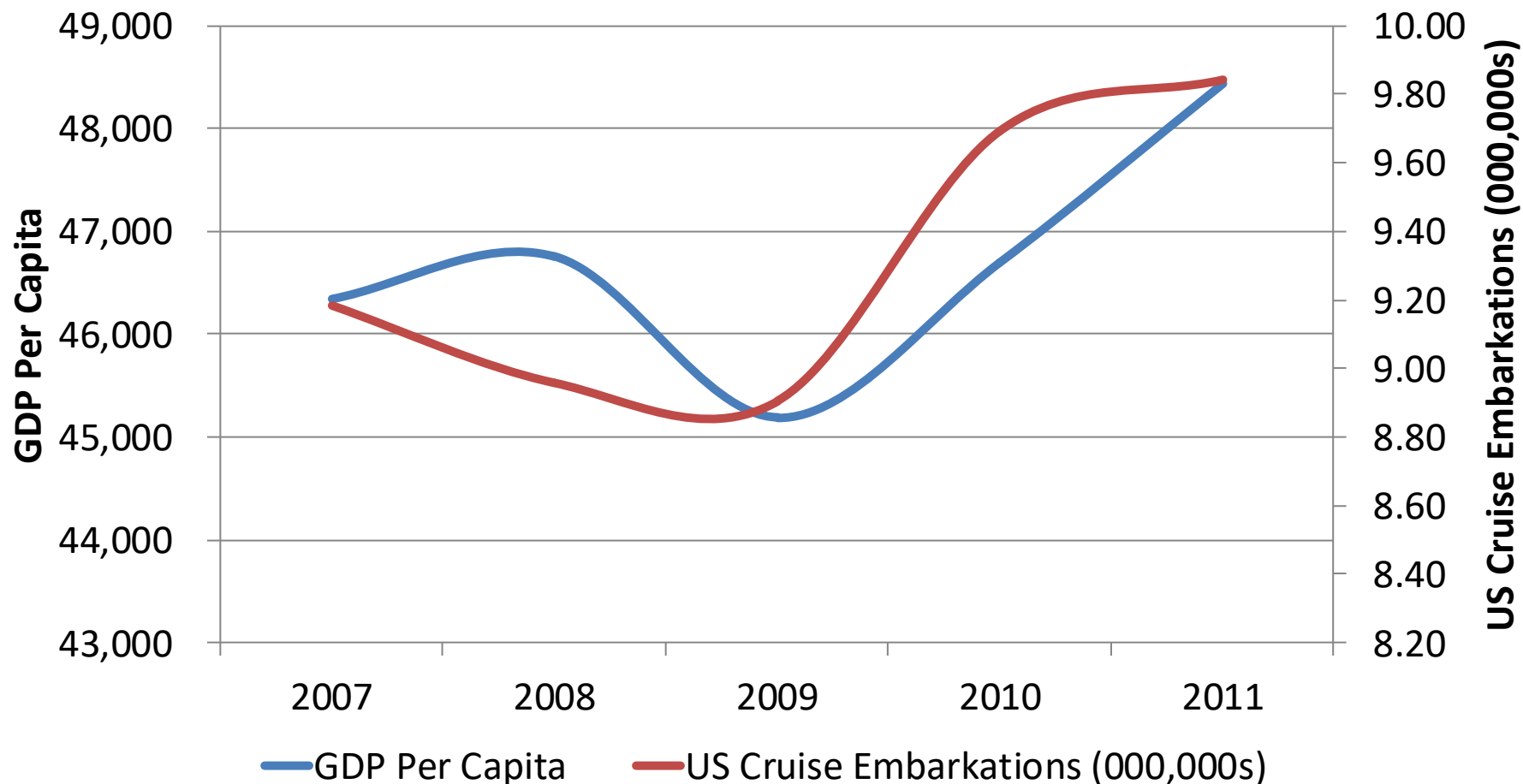
Income Elasticity

- How about cruise ship vacations?



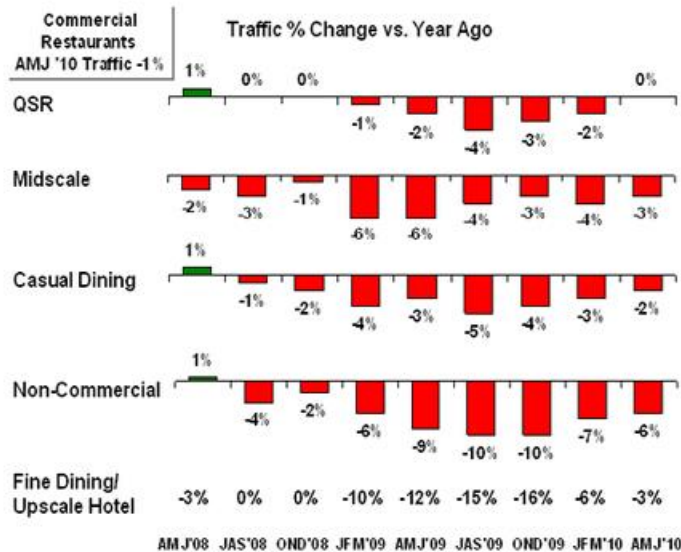
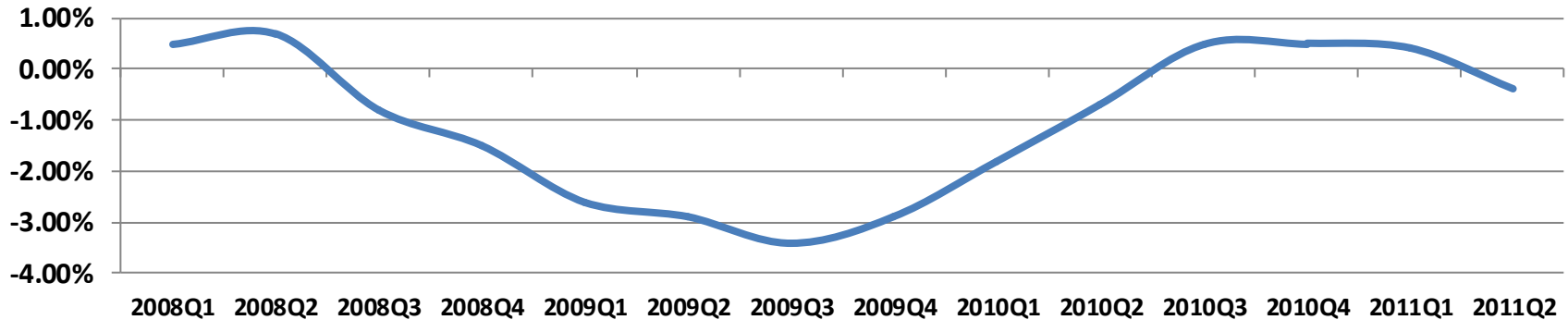
Income Elasticity

- What is the income elasticity of cruise trips?
 - The income elasticity is positive (cruises are a normal good)



Substitution Within Category?

Year-Over-Year Change in Restaurant Traffic



Source: The NPD Group / CREST

- Consumers started eating out less overall
- However, “QSR” (Quick Service Restaurants) saw lower decreases than higher-end options.

Summary and Direction

Today

Own-price elasticity: negative. Inelastic if between 0 and -1, elastic if < -1 .

Cross-price elasticity: negative for complements, positive for substitutes.

Income elasticity: inferior goods get consumed less as income rises; normal goods more, and luxury goods far more.

Break Question: Can you give me an estimate of an “elasticity” in your life?
(remember: just a % change in one variable with respect to a % change in the other)

Next Lecture

Government interventions:

- Who “pays” for taxes?
- When might suppliers favor an intervention like price regulation?
- Why do governments intervene? Externalities.